

ASSIGNMENT SPECIFICATION

CLO1: Use appropriate design and evaluation techniques required in the development of usable interfaces. (P4, PLO3)

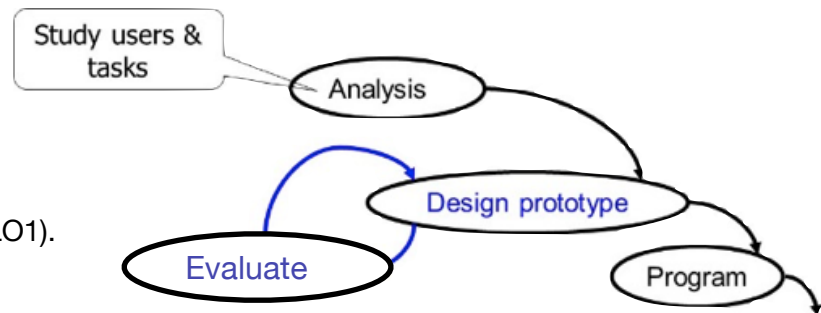
CLO2: Evaluate the impact of good and bad interfaces on the usability of applications and the stakeholders. (C4, PLO2)

1) Assignment Objective:

-Learn the HCI design process.

2) Assignment Mode:

- Group assignment.
- 3-5 students per group.
(Approval from tutor for >5 students)
- 4 reports and presentation (CLO1).
 - (i) Proposal (CLO1)
 - (ii) Analysis Report (CLO1)
 - (iii) Design & Prototyping Report (CLO1)
 - (iv) Evaluation Report (CLO2)



3) Assignment Tasks:

Select a computer system with software interface, and develop a prototype.

Week	Tasks/Activities
W1	- Form assignment group. - Each group proposed a computer system with software interface and get approval from your tutor. Hand in group name list and assignment title. - Preparing proposal. [Refer to Appendix A]
W2	- Users analysis with Personas. - Task analysis with Hierarchical Task Analysis (HTA), Scenario of current task, Environment analysis. Hand in Proposal (soft copy with cover page)
W3	- Preliminary design with Storyboard. Hand in Analysis Report (soft copy) [Refer to Appendix B]
W4	Prototyping
W5	- Writing Test tasks - Heuristic Evaluation & Usability Testing [Refer to Appendix F] Hand in Design & Prototyping Report (soft copy) [Refer to Appendix C]
W6	- Presentation [Refer to Appendix E] Hand in Evaluation Report (soft copy) [Refer to Appendix D]
W7	Assignment feedback.

4) Report Specifications :

- Font type: Times New Roman or Arial
- Font size: 11 or 12 (use 14 or slightly bigger size for title)
- Line spacing: Single/1.5

5) Late Policy:

- ALL assignment deliverables must be submitted on due date.
- For late submission, there will be a reduction of absolute marks from the mark's score submitted:
 - Late 1 to 3 days after deadline of submission: minus 10%;
 - Late 4 to 7 days after deadline of submission: minus 20%;
 - Late more than 7 days after deadline of submission: minus 100%

=====APPENDIX A=====**Proposal (20%) -CLO1**

Proposal should included all of the following parts:

(i) Introduction (5 marks)

- Brief description of your proposed computer system, target audiences/users.

(ii) PACT Analysis (8 marks)

- People, Activities, Contexts, Technologies (PACT) Analysis.

(iii) Usability goals (5 marks)

- Choose and justify the usability goals which you consider to be the important goals for your system.

(iv) Reference (2 marks)

- List your references at this part, and use at least 1 academic reference in this report.

=====APPENDIX B=====**Analysis Report (30%)-CLO1**

Analysis Report should included all of the following parts:

(i) Persona (10 marks)**(ii) Task Analysis of current task (10 marks)****(iii) Scenario of current task (5 marks)****(iv) Environment analysis (5 marks)**

**** Do not write about your system design, it does not exist at this analysis stage.**

=====APPENDIX C=====**Design & Prototyping Report (20%)- CLO1**

Design & Prototyping Report should included all of the following parts:

(i) Preliminary Design (5 marks)

- Storyboard.

(ii) Detail Design (15 marks)

- Screen shot of your high fidelity prototype to show your user interface design (5 marks).

**** you are encourage to refer to the good application/system/software/website for their UI design.**

- Explain your design decisions from the aspects of : user needs, usability goals and cognitive issues. (10 marks)

**** This part requires reasoning behind your design decisions. Explanation of design decision without any reasoning will not given any mark.**

=====APPENDIX D=====**Evaluation Report (20%)- CLO2**

- Nielsen's Heuristic Evaluation

=====APPENDIX E=====

Presentation (10%)- CLO1

- In this presentation, you are expected to show the knowledge/skills you have learned from this course.

=====APPENDIX F=====

Brief guidelines for conducting usability testing

Before starting:

- Introduce your prototype to the tester
- Acknowledge that the prototype may have usability problems
- Ask for permission to record the usability test

[Examples of software for recording everything on the computer screen and save it as video files
(1) www.bandicam.com (2) <http://screencast-o-matic.com/>]

During testing:

- Hand out test task one at a time to the tester
- Remember to ask the tester to "think aloud"
- Observe the tester interacting with your prototype. Let the tester perform his/her tasks independently.
- Observe the tester all the time during testing and record problems faced by the tester (E.g. long paused, excessive eyes movement, thinking, sighing, annoying, etc)

After testing:

- Thanks the testers for his/her time
- Asking the testers for their opinions about your prototype.